

SYSTEMATIC REVIEW

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Effects of different interventions on anxiety disorders in children and adolescents: a systematic review and bayesian network meta-analysis

Longhui Li¹, Qiner Li¹, Jingyi Wang¹, Quan Fu^{1*} and Meng Chi^{2*}

Abstract

Background Anxiety disorders are among the most prevalent psychiatric conditions, particularly affecting children and adolescents. Since the COVID-19 pandemic, the incidence of anxiety in this population has increased, making it a significant public health concern. This study aimed to evaluate the effectiveness of various interventions for anxiety disorders in children and adolescents and identify the most effective approach to mitigate the impact of these disorders.

Methods We conducted a thorough search of eligible randomized controlled trials across five databases: Cochrane, Embase, PubMed, Scopus, and Web of Science. A Bayesian network meta-analysis was performed using R Studio, and the quality of evidence was evaluated using the GRADE method.

Results From 19,442 publications retrieved, 30 RCTs involving 1,711 participants were included. The results showed that Acceptance and Commitment Therapy was the most effective intervention for treating anxiety disorders in children and adolescents (MD = -3.83 [95% CrI: -9.33, 1.51]). Cognitive Behavioral Therapy was the second most effective (MD = -3.64 [95% CrI: -7.36, -0.48]), followed by Virtual Reality Exposure Therapy (MD = -2.53 [95% CrI: -8.23, 3.32]) and Physical Exercise (MD = -2.16 [95% CrI: -9.99, 5.52]).

Conclusions Acceptance and Commitment Therapy (ACT) appears to be the most effective intervention for anxiety disorders in children and adolescents. However, this finding should be interpreted with caution due to the overall low quality of evidence, high heterogeneity, and imprecision in the results. Future research should investigate the potential benefits of combining physical exercise or virtual reality-assisted therapies. Moreover, large-scale, high-quality randomized controlled trials are essential to further validate and refine these findings.

Trial registration PROSPERO, CRD42024587910.

Keywords Anxiety, Psychotherapy, Physical exercise, Virtual reality, Children, Adolescent, Network meta-analysis

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Introduction

Anxiety disorders are among the most prevalent psychiatric conditions, encompassing diagnoses such as specific phobias (e.g., fear of needles, spiders), agoraphobia, social phobia, generalized anxiety disorder, panic disorder, and separation anxiety disorder [1]. Globally, an estimated 301 million people—approximately 4% of the population—are affected by these disorders, making them the most prevalent psychiatric conditions [2]. Adolescents are particularly vulnerable. Around 4.4% of children aged 10–14 and 5.5% of those aged 15–19 are estimated to suffer from an anxiety disorder [3]. Anxiety disorders typically develop before the age of 14 [4] and are linked to a range of short- and long-term consequences, including diminished quality of life, increased psychiatric and physical comorbidities, disability, school dropout, and, in severe cases, suicide and reduced life expectancy [5, 6]. The impact of anxiety disorders has intensified following the COVID-19 pandemic, with elevated rates of anxiety among children and adolescents attributed to factors such as social isolation, school closures, and reduced physical activity [7–10]. As a result, psychological issues in this age group have become a major public health concern [11]. Unfortunately, the remission rates following treatment remain low, with recovery rates in adolescents hovering around 50% [12]. Given these challenges, identifying effective interventions to improve outcomes, reduce relapse, and enhance recovery is crucial.

Current treatments for anxiety disorders in children and adolescents typically include medication and psychotherapy. Pharmacological treatments, such as selective serotonin reuptake inhibitors (SSRIs) and serotonin-norepinephrine reuptake inhibitors (SNRIs), have shown effectiveness in reducing symptoms [13]. However, these medications are often associated with side effects, including nausea, dizziness, headaches, sleep disturbances, and gastrointestinal issues [13, 14]. In some cases, these side effects may even exacerbate anxiety symptoms and increased risk of suicide [13]. Medication adherence also poses a challenge, as many young patients and their families are hesitant to commit to long-term use, which can limit the overall effectiveness of pharmacological approaches [14, 15]. In contrast, psychotherapy is widely regarded as a safer and more effective treatment option, offering sustained benefits, fewer side effects, greater acceptability, and strong potential for future application [16–18]. Given these factors, our study focuses on evaluating non-pharmacological psychological interventions.

Among the various forms of psychotherapy, Cognitive Behavioral Therapy (CBT) is considered the gold standard, particularly for adolescents [19]. CBT focuses on identifying and modifying maladaptive thinking and behavior patterns that contribute to anxiety. It typically involves techniques such as exposure to anxiety-inducing

situations, cognitive restructuring of negative thoughts, relaxation strategies, and positive self-talk [20, 21]. Numerous studies have demonstrated the clinical benefits of CBT for adolescent anxiety disorders [22–24]. Another widely studied intervention Acceptance and Commitment Therapy (ACT), a modern variation of CBT. ACT emphasizes psychological flexibility—the ability to adapt to difficult thoughts and emotions while acting in line with personal values [25]. It is built around six core processes: acceptance, cognitive defusion, present-moment awareness, the self as context, values, and committed action [25, 26]. Unlike traditional CBT, which aims to modify distressing thoughts, ACT encourages individuals to observe and accept them without judgment [25, 27]. By fostering a more detached and accepting mindset, ACT helps patients see their anxious thoughts as separate from their core identity, reducing the impact of these thoughts on their overall well-being [27]. Research suggests that ACT is also effective in reducing anxiety symptoms among adolescents [28–30].

Beyond cognitive-behavioral approaches, technology-based exposures—such as virtual reality exposure therapy (VRET)—are increasingly recognized as effective tools for exposure therapy in treating anxiety disorders. Virtual reality (VR) involves a computer-generated environment that simulates real-life scenarios, allowing users to experience immersive, interactive settings [31]. These environments often incorporate sensory feedback—such as visual, auditory, and tactile cues—to create the feeling of being physically present in a digital world [32, 33]. VRET makes use of this immersive technology to gradually expose individuals to anxiety-provoking situations in a safe and controlled virtual environment [34]. This method closely mirrors the principles of real-life exposure therapy but offers greater flexibility and control over the intensity and pacing of exposure [35]. Several studies have shown that VRET is effective in treating various anxiety disorders in adolescents, including public speaking anxiety [35–37], social anxiety disorder [38], school anxiety [39, 40], and specific phobias [41, 42].

While psychotherapy is a common treatment for anxiety disorders, it often comes with barriers such as social stigma and high costs [43]. Moreover, traditional psychological therapies typically do not address the physical health challenges that often accompany anxiety disorders [44]. These limitations have prompted growing interest in physical exercise (PE) as a complementary or alternative approach [45, 46]. Numerous studies have explored the impact of PE on adolescent anxiety disorders, with many suggesting that its effects are comparable to those of psychotherapy and medication [47–49]. Although the mechanisms underlying the benefits of PE on anxiety are not yet fully understood, several neurochemical pathways have been proposed. PE increases the release

of β -endorphins, which can improve mood and reduce anxiety [50]. It also supports the regulation of the hypothalamic-pituitary-adrenal (HPA) axis—a system often disrupted in individuals with anxiety disorders—thereby helping to stabilize stress-related hormones such as cortisol and catecholamines [51, 52]. In addition, physical activity has been shown to boost levels of brain-derived neurotrophic factor (BDNF), a protein vital for brain function and emotional regulation [53]. Since reduced BDNF levels are commonly observed in those with anxiety, its increase through regular exercise may play a key role in alleviating symptoms [54, 55].

While ACT, CBT, PE, and VRET all show promise, the evidence is scattered. Most previous reviews focus on only one or two treatment types and often rely on traditional meta-analysis that limit comparisons to treatments tested directly against one another [22, 24, 56]. This makes it difficult to determine which interventions are most effective across the broader landscape. To address this gap, we conducted a Bayesian network meta-analysis, a statistical approach that allows for comparisons across multiple interventions—even if they have not been directly compared in individual studies [57]. This method also allows for the integration of prior information and performs well with smaller samples, offering a more flexible and comprehensive approach to evidence synthesis [58].

Given the rising prevalence and earlier onset of anxiety disorders in children and adolescents, identifying the most effective interventions is essential. Therefore, this study aimed to compare and rank the effectiveness of ACT, CBT, PE, and VRET in treating anxiety disorders in this population using Bayesian network meta-analysis.

Materials and methods

Protocol and registration

This review was conducted in line with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines [59], with a full checklist available in Appendix A. The protocol was registered on September 21, 2024, with the International Prospective Register of Systematic Reviews (PROSPERO) under the ID CRD42024587910. Further details can be accessed at <https://www.crd.york.ac.uk/prospero/>.

Search strategy and selection criteria

We conducted a comprehensive search of five major databases—Cochrane Library, Embase, PubMed, Scopus, and Web of Science—for studies published between January 1, 1976, and September 1, 2024. To ensure completeness, we also screened Google Scholar and reviewed the reference lists of relevant systematic reviews to identify additional eligible studies. Only peer-reviewed articles published in English were included. We excluded gray

literature, unpublished studies, and entries from clinical trial registries to maintain consistency in data quality and reporting standards. The search and screening process was independently carried out by two reviewers (LLH and WJY), following the recommendations of the Cochrane Handbook for Systematic Reviews [60]. Any disagreements were resolved through discussion with a third reviewer (LQE). The full search strategy is detailed in Supplementary Table 1.

Eligible studies were selected based on the PICOS (Population, Intervention, Comparison, Outcome, and Study design) framework. Inclusion and exclusion criteria were as follows: (1) Population: This study included children and adolescents aged 6 to 18 years diagnosed with anxiety disorders according to DSM-5 criteria, including specific phobias, agoraphobia, social anxiety disorder, generalized anxiety disorder, panic disorder, and separation anxiety disorder. To reduce clinical and methodological heterogeneity, individuals with major comorbid conditions—such as autism spectrum disorder (ASD), Attention-deficit/hyperactivity disorder (ADHD), cancer, or diabetes—were excluded. These conditions often require substantial adaptations to psychological treatments, which can influence both the structure and effectiveness of interventions [24]. For instance, cognitive-behavioral therapy for children with ASD frequently incorporates tailored components such as social stories, visual supports, and structured worksheets [61, 62]. (2) Intervention: Included studies evaluated one or more of the following interventions: cognitive behavioral Therapy (CBT), acceptance and commitment therapy (ACT), physical exercise (PE), and virtual reality exposure therapy (VRET). Studies combining two of these interventions—such as CBT combined with physical exercise—were also considered. (3) Control group: Eligible control conditions included usual care, no intervention, or a wait-list group. (4) Outcome: The primary outcome of interest was anxiety symptoms in children and adolescents. Studies with unclear results, incomplete data or data that could not be reliably extracted were excluded. (5) Study design: Only randomized controlled trials were included. Non-randomized studies, longitudinal observational research, dissertations, and conference abstracts were excluded.

Study selection

All references for the studies included in this review were managed using EndNote X9. After removing duplicates, two reviewers (LLH and WJY) independently screened the remaining records. The initial screening was based on titles and abstracts to identify potentially eligible studies, followed by a full-text review to determine final inclusion. At each stage, consensus between the two reviewers was required. Any disagreements were resolved through

discussion, with a third reviewer (LQE) consulted when necessary. To assess inter-reviewer reliability, 30% of the full-text articles were randomly selected for independent evaluation by both reviewers. Agreement was measured using the kappa statistic.

Data extraction

Data extraction was done independently by two researchers (LLH, WJY). Extracted data from all included studies were then compared and discrepancies, if any, were resolved by a third reviewer (LQE). Extracted data included authors' names, year of publication, basic information about the study population (country, age, gender, social status, recruitment), characteristics of the intervention (means of intervention, duration, frequency, time of intervention, time of follow-up), outcome and measurement instruments. When required information was missing, it was recorded as not reported (NA).

Assessment of quality of individual studies

The risk of bias was assessed for each study using Cochrane Review Manager 5.4, based on the methodological quality assessment criteria outlined in the Cochrane Handbook 5.1.0. Included studies were assessed according to six indicators, including method of random allocation, allocation concealment, blinding, completeness of outcome data, selective reporting of study results and other sources of bias. The veracity of the reporting of study results and any other potential sources of bias were also assessed. Risk of bias was assessed qualitatively based on descriptions provided in the included studies, and each indicator was categorized as having a high, low or uncertain risk of bias.

Data Preparation

Before conducting the meta-analysis, all outcome measures and scales from the included studies were standardized. In studies with multiple intervention or control groups, pooled means and standard deviations were calculated for the relevant intervention groups prior to estimating effect sizes [63]. When multiple related scales were used to assess the same outcome, the primary measure was prioritized. If no primary measure was identified, the means and standard deviations of the relevant scales were pooled to derive a composite estimate.

Statistical analysis

We performed a Bayesian network meta-analysis using the Multinma package (version 0.8.1) [64] in RStudio (version 2024.12). The analysis used 4 MCMC chains, each with 2000 iterations, of which 1000 were burn-in and 1000 valid samples were retained per chain. Vague prior distributions were applied to intercepts, treatment effects, and heterogeneity parameters. Effect sizes were

measured using the mean difference (MD) and 95% credible intervals (95%CrI), accounting for continuous outcomes on different scales. The MD represents the average difference in outcomes between groups, while the CrI reflects the range within which the true effect is likely to lie with 95% probability. Model selection was based on several fitting parameters, including the deviance information criterion (DIC), residual bias relative to the data points, the number of parameters (PD), and the contribution of residual bias for each data point. Heterogeneity was evaluated by comparing fixed and random effects models and reporting the between-study standard deviation (τ). Further analyses included assessing the relative effectiveness of the interventions versus control, as well as determining treatment rankings, ranking probabilities, cumulative ranking probabilities.

Network visualization was used to display the structure of the evidence and show the relationships between studies [65]. The ranking probabilities and cumulative ranking probability was summarized using surfaces under the cumulative ranking area (SUCRA). The SUCRA values, which range from 0 to 100%, reflect the likelihood of each treatment achieving the best effect, with higher values indicating better treatment outcomes. To assess the global consistency of the network, we compared the posterior mean residual bias, the deviance information criterion (DIC), and the inter-study standard deviation (SD) between consistent and inconsistent models [66]. Local consistency was evaluated using a node-splitting approach. The certainty of evidence for each comparison was assessed using the GRADE approach adapted for network meta-analysis [67]. Unlike traditional pairwise meta-analyses, network meta-analyses include studies comparing different sets of interventions, requiring a method to meaningfully rank treatments [68]. In this study, a comparison-adjusted funnel plot and comparison-adjusted Egger's test were used to assess potential small-study effects, with treatments ordered by effect size from largest to smallest.

Result

Literature screening process and results

Following the initial database search, 19,442 records were identified, with an additional 6 studies retrieved through manual searching. After removing duplicates using End-Note, 10,596 records remained. Title and abstract screening reduced this number to 79 potentially eligible studies. Full-text assessment resulted in 30 studies meeting the inclusion criteria for the meta-analysis. Inter-reviewer reliability during the full-text screening stage was high, with a Kappa statistic of 0.83. A detailed summary of the study selection process, including excluded studies and reasons for exclusion, is presented in Fig. 1 and Supplementary Table 2.

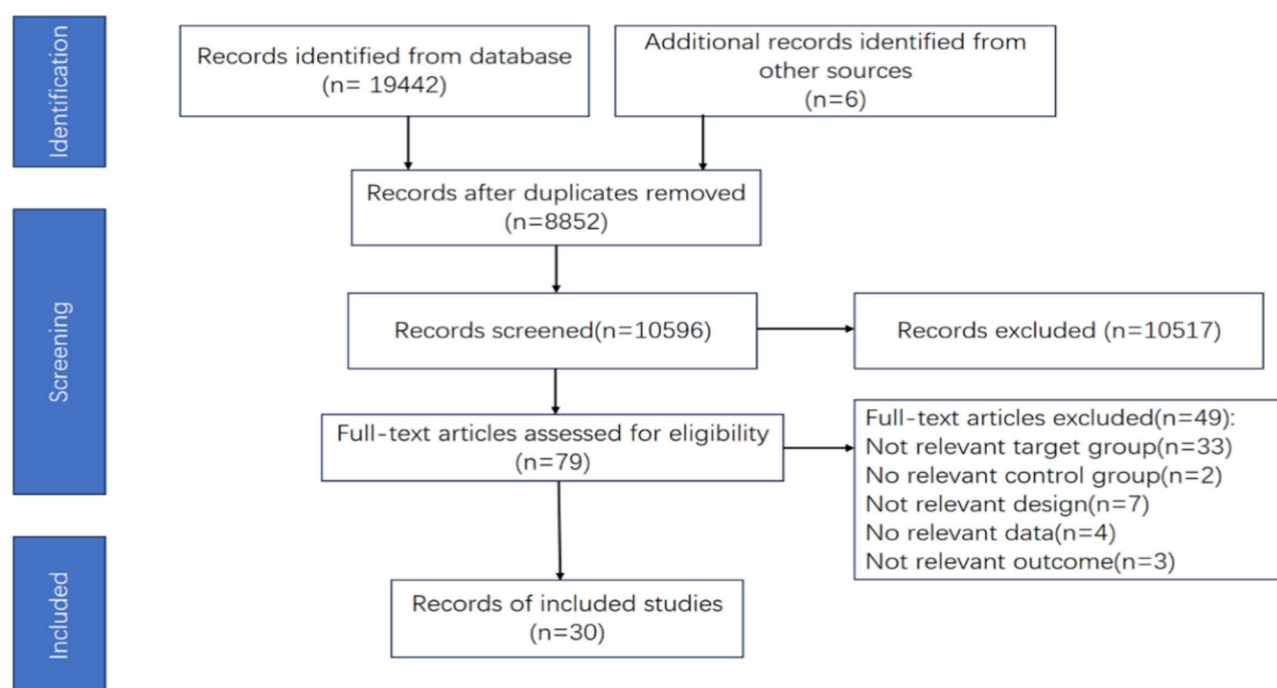


Fig. 1 PRISMA flow chart

Study characteristics

The network meta-analysis included 30 studies with 1,711 unique participants. Detailed study characteristics are provided in the Table 1. These studies originated from 12 countries, the United States ($n=7$), Australia ($n=5$), Germany ($n=3$), the United Kingdom ($n=2$), Iran ($n=2$), Sweden ($n=2$) and other regions ($n=9$). Most studies did not report participants' social status, with recruitment primarily from hospitals ($n=13$), schools ($n=13$), and communities ($n=4$). The average participant age across 27 studies was 12.40 years, and 62% of participants from 29 studies were female.

The analysis evaluated four interventions ACT, CBT, PE, and VR on adolescents with anxiety disorders. Specifically, the interventions were distributed as follows: ACT($n=6$), CBT ($n=16$), PE ($n=2$), VRET ($n=6$). The median intervention duration was 10 weeks (range: 3–24 weeks), with sessions typically held once a week (range: 1–5 times) and lasting 60 min (range: 20–120 min). The median follow-up duration was 3 months (range: 1–12 months). In all studies, anxiety symptoms were assessed using self-reported measures. The three most commonly used scales were the Spence Children's Anxiety Scale, the Multidimensional Anxiety Scale for Children, and the Public Speaking Anxiety Scale, which were utilized in seven, five, and two studies, respectively.

NA: Not Reported; ADIS: Anxiety Disorders Interview Schedule; FSSC-R: Fear Survey Schedule for Children Revised; HADS: Hospital Anxiety & Depression Scale; HAMA: Hamilton Anxiety Scale/Hamilton Depression

Scale; LSAS-CA-SR: Liebowitz Social Anxiety Scale for Children and Adolescents- Self-report Version; MASC: Multidimensional Anxiety Scale for Children; PSAS: Public speaking anxiety symptoms; RCADS: Revised Children's Anxiety and Depression Scale; SAD: Social Anxiety Scale; SAI: Separation Anxiety Inventory for Children; SASC: Social Anxiety Scale for Children; BAI: Beck Anxiety Inventory; SCARED: Screen for Child Anxiety Related Disorders; SCAS: Spence Children's Anxiety Scale; SCAS-S: Spence children's anxiety scale-short version; SMA: Severity measure for acrophobia; SPAIK: Social Phobia and Anxiety Inventory for Children, German version; SPQC: Spider Phobia Questionnaire for Children; STAIC-T: State-Trait Anxiety Inventory for Children–Trait Version; VAS: Visual analogue scales.

Results of research quality assessment

The risk of bias assessment for the 30 included studies is summarized in Fig. 2. Of these, 13 studies demonstrated a low risk of bias in random sequence generation, while one study showed high bias due to non-randomization. The allocation method was not clearly described in 23 studies, 7 studies concealed allocation adequately. Most studies, however, did not provide detailed descriptions.

Double-blinding was rarely mentioned; only 2 studies blinded patients, 6 blinded outcome evaluators, and 2 had a high risk of bias as participants were not blinded. Data completeness was generally good in 23 studies. 5 studies lacked sufficient information to assess the reasons for missing data, and 2 were at high risk due to unequal

Table 1 Characteristics of included studies

Author	Country	Social Status	Age/Women Percent-age/Sample size (intervention/control)	Intervention	Duration(week)/Frequency (times per week/min)	Follow up(month)	Measure-ment scales
Azimisefat et al. 2022 [69]	Germany	NA	17.2;100%;15/15	VRET	6;2;60	6	SMA
Gallagher et al. 2004 [70]	American	NA	8–11;52%;12/11	CBT	3;1;180	1	MASC
Gutiérrez-Maldonado et al. 2009 [40]	Barcelona	NA	11.9;64%;18/18	VRET	6;1;45	NA	SASC
Hancock et al. 2018 [71]	Australia	Mix	11;58%;54/46	ACT/CBT	10;1;90	3	FSSC-R
Hudson et al. 2009 [72]	Australia	NA	10.2;43%;52/46	CBT	10;1; NA	6	SCAS
Infantino et al. 2016 [73]	Japan	NA	10.9;57%;25/24	CBT	10;1;60–120	6	SCAS
Ishikawa et al. 2019 [74]	Japan	NA	10.9;57%;25/24	CBT	10;1;60–120	6	SCAS
Kahlon et al. 2023 [36]	American	NA	14.16;84%;32/17	VRET	3;1;30–60	NA	PSAS
Kahlon et al. 2024 [37]	American	NA	14.2;84%;16/15	VRET	3;3;30	3	PSAS
Leigh and Clark 2023 [75]	England	Mix	16.22;91%;22/21	CBT	14;1;20	6	LSAS-CA-SR
Melfsen et al. 2011 [76]	Germany	NA	10.68;47%;21/23	CBT	24;1;50	NA	SPAIK
Nissling et al. 2023 [77]	Sweden	NA	17.3;81%;23/19	ACT	10; NA; NA	NA	SCAS-S
Parker et al. 2016 [78]	Australia	low	17.6;61%;88/86	PE	6;5;60	NA	BAI
Petersen et al. 2023 [79]	American	low	15.7;73%;12/9	ACT	12;1;60	1	SCARED
Philippot et al. 2022 [49]	Belgium	NA	15.35;63%;20/20	PE	5;4;60	NA	HADS
Pincus et al. 2010 [80]	American	Mix	15.75;73%;13/12	CBT	12;1;50	6	MASC
Rostami et al. 2014 [81]	Iran	NA	12–16;0%;20/20	ACT	10;1;60	3	SAD
Schneider et al. 2011 [82]	Germany	NA	6.24; NA;14/17	CBT	12;2;50	1	SAI
Smith et al. 2014 [83]	American	Mix	9.8;39%;14/12	CBT	10;1;60	3	MASC
Southam-Gerow et al. 2010 [84]	American	Mix	10.9;56%;17/19	CBT	16;1; NA	NA	STAIC-T
St-Jacques et al. 2010 [85]	Canada	NA	10.16;84%;17/14	VRET	6;4;60	6	SPQC
Stjerneklar et al. 2019 [86]	Denmark	NA	15.03;70%;32/31	CBT	14;1;30	12	SCAS
Sülter et al. 2022 [87]	Netherlands	NA	10.46;49%;36/29	VRET	3;1;30	NA	VAS
Swain et al. 2015 [88]	Australia	NA	13.8;63%;16/23	ACT/CBT	10;1;90	3	ADIS
Thirlwall et al. 2013 [89]	England	NA	7–12;48%;87/57	CBT	8;1;40	6	SCAS
Vigerland et al. 2016 [90]	Sweden	NA	10.1;55%;45/45	CBT	10;1; NA	3	SCAS
Waters et al. 2009 [91]	Australia	NA	6.76;53%;11/11	CBT	10;1;60	12	ADIS
Wergeland et al. 2014 [92]	Norway	Mix	11.5;53%;144/38	CBT	10; NA; NA	12	SCAS
Yen et al. 2014 [93]	China	NA	9.1;50%;18/26	CBT	17;1; NA	NA	MASC
Zemestani et al. 2022 [94]	Iran	low	15.2;100%;24/45	ACT	8;1;45–60	1	RCADS

amounts and reasons for missing data across intervention groups.

Selective reporting was detected in one study, indicating high bias, while 27 studies showed no signs of it. For the remainder, there was insufficient information for assessment. 11 studies revealed no other biases, but for the others, other biases remain uncertain. Detailed risk of bias information is provided in the Supplementary Fig. 1.

Results of network meta-analysis

Network geometry

The network plot is shown in Fig. 3, where our network consists of 5 nodes (i.e., interventions) with 5 pairs of direct comparisons and 1 closed loop. The most common head-to-head comparison was the CBT intervention versus the control group ($n = 16$).

Model selection, heterogeneity and consistency tests

For model selection, the random effects model demonstrated better fit, with a deviance information criterion (DIC) of 141.3 and a residual deviance of 86.7. In comparison, the fixed effects model yielded a higher DIC of 263.4 and a residual deviance of 229.3. The posterior mean residual deviance of the random effects model closely matched the number of data points, further supporting its suitability. The random effects model also indicated substantial heterogeneity ($\tau = 5.87$; 95% CrI: 2.92, 9.83), which may reflect variations in study design, participant characteristics, or intervention delivery across studies. These findings support the appropriateness of using the random effects model. DIC distribution plots for both models are presented in Supplementary Figs. 2 and 3, and

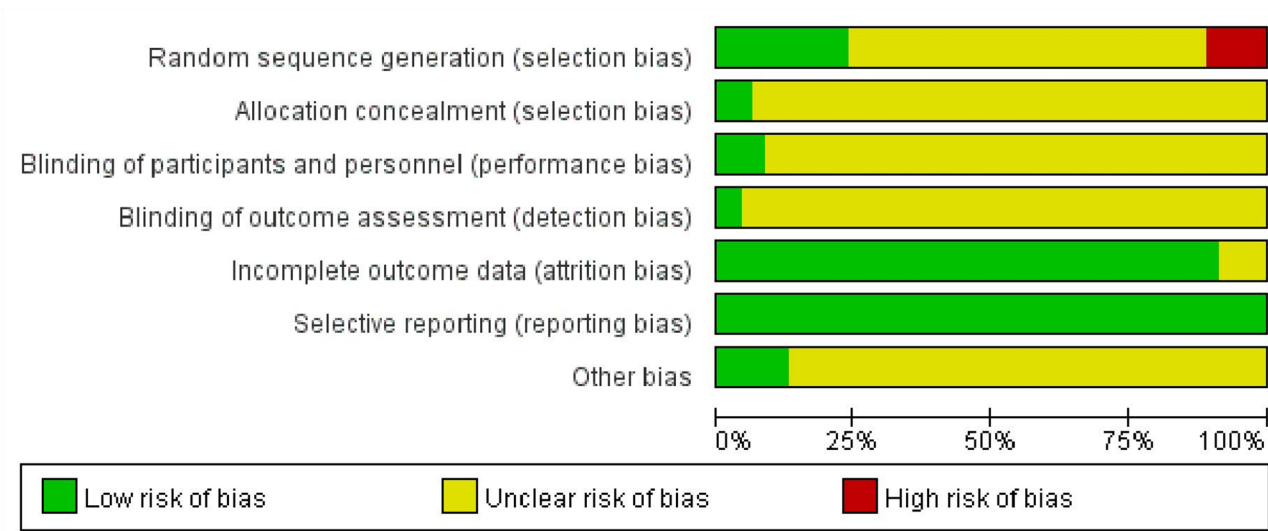


Fig. 2 Literature quality assessment

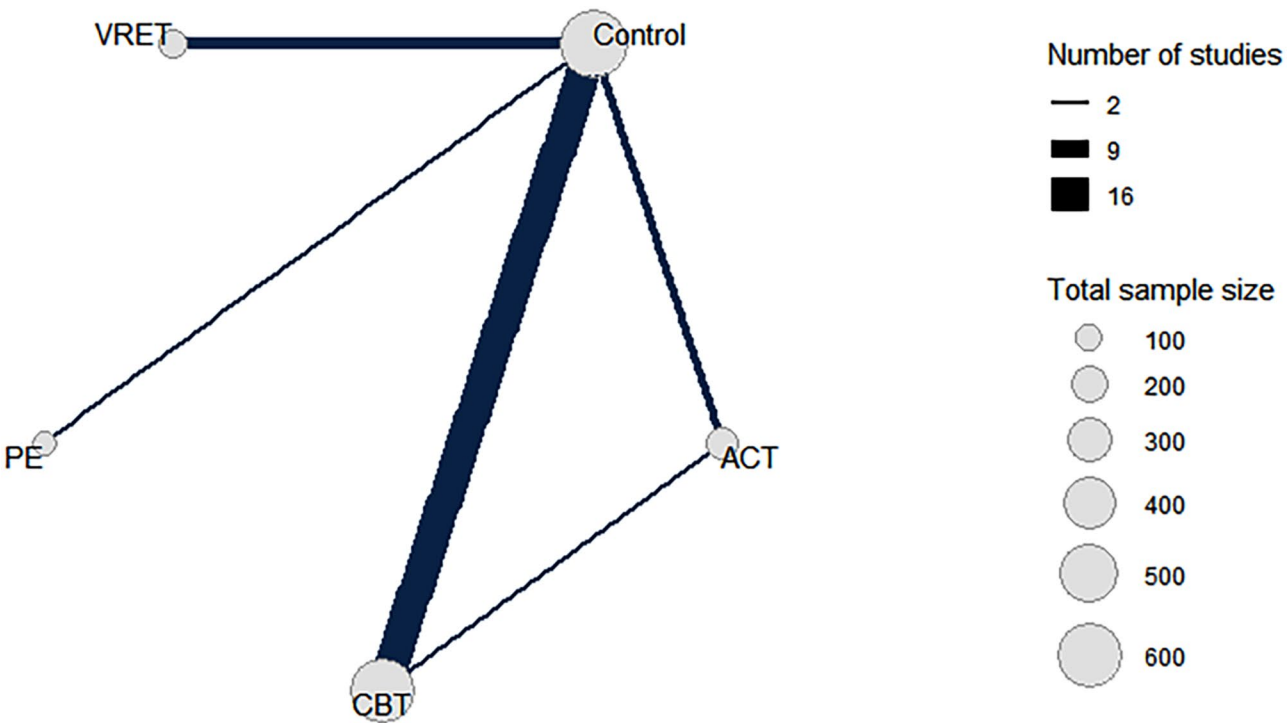


Fig. 3 Network plot of comparisons

the posterior distribution plots for the selected model are shown in Supplementary Fig. 4.

For the consistency test, the DIC and residual deviance were 141.3 and 86.7 for the consistent model, and 142.6 and 87.5 for the inconsistent model, respectively. These values suggest no significant inconsistency among the studies. Additionally, the results from the node-splitting method showed all P -values > 0.05 , indicating no

evidence of local inconsistency. The local inconsistency plots can be found in Supplementary Fig. 5.

Relative effects, ranking of effectiveness and SUCRA

The ACT intervention was the most effective intervention for adolescents with anxiety disorders, with a MD of -3.83 [95% CrI: $-9.33, 1.51$]. CBT was the second most effective (MD = -3.64 [95% CrI: $-7.36, -0.48$]), followed by VRET (MD = -2.53 [95% CrI: $-8.23, 3.32$]) and PE

(MD = -2.16 [95% CrI: -9.99, 5.52]). The relative effects of these interventions are illustrated in Fig. 4.

Effectiveness ranking and SUCRA rankings indicated that ACT was the most likely to be the most effective intervention for this population, with a mean ranking of 2.25 [95% CrI: 1, 5] and a SUCRA value of 0.69. CBT followed closely, with a mean ranking of 2.31 [95% CrI: 1, 4] and a SUCRA value of 0.66. PE and VRET both had mean rankings of 3.12 [95% CrI: 1, 5] and 2.86 [95% CrI: 1, 5], respectively, with SUCRA values of 0.51 for each. The SUCRA plot and ranking probabilities plot are presented in Fig. 5 and Supplementary Fig. 6.

Certainty of evidence and publication bias

The GRADE system defines the overall certainty of the evidence included in this meta-analysis as low due to the presence of publication bias and gross imprecision (i.e., 95% CrI of zero, with a wide CrI indicating uncertainty in the estimates). Supplementary Table 3 provides the quality of evidence in each of the direct, indirect, and network meta-analysis estimates.

We assessed publication bias using a funnel plot. As shown in Fig. 6, some of the findings in the funnel plot were asymmetrically distributed, so we further performed the Egger test to quantify the publication bias test. Egger's test showed $p = 0.91$, indicating symmetry in the funnel plot, which confirms that there are no small-study effects.

Discussion

This study aimed to identify the most effective interventions for treating anxiety disorders in children and adolescents, while also comparing the effectiveness of ACT, CBT, PE, and VR. After analyzing 30 randomized controlled trials, we confirmed that all four interventions—ACT, CBT, PE, and VRET—were effective in treating anxiety disorders in this population. Among them, ACT emerged as the most effective, followed by CBT, PE, and VRET.

This study found that psychotherapy interventions, particularly ACT, are the most effective treatments for anxiety disorders in children and adolescents, with ACT outperforming CBT. This finding contrasts with the results of Zhou et al. [95] and In-Albon and Schneider [22], who suggested that CBT might be the preferred choice for treating anxiety disorders in this population. However, their studies focused solely on CBT and did not include ACT, PE, and VRET. Our study refines previous research by providing a more comprehensive and up-to-date comparison of these interventions. The primary goal of ACT is to enhance psychological flexibility, which is closely linked to improved mental health [96]. Lower psychological flexibility is associated with higher levels of anxiety, depression, low self-esteem, and rumination, making it a critical factor in treating anxiety disorders [97]. A recent meta-analysis also confirmed the effectiveness of ACT in reducing anxiety symptoms in adolescents, particularly in cases characterized by psychological inflexibility [98]. Furthermore, our study also

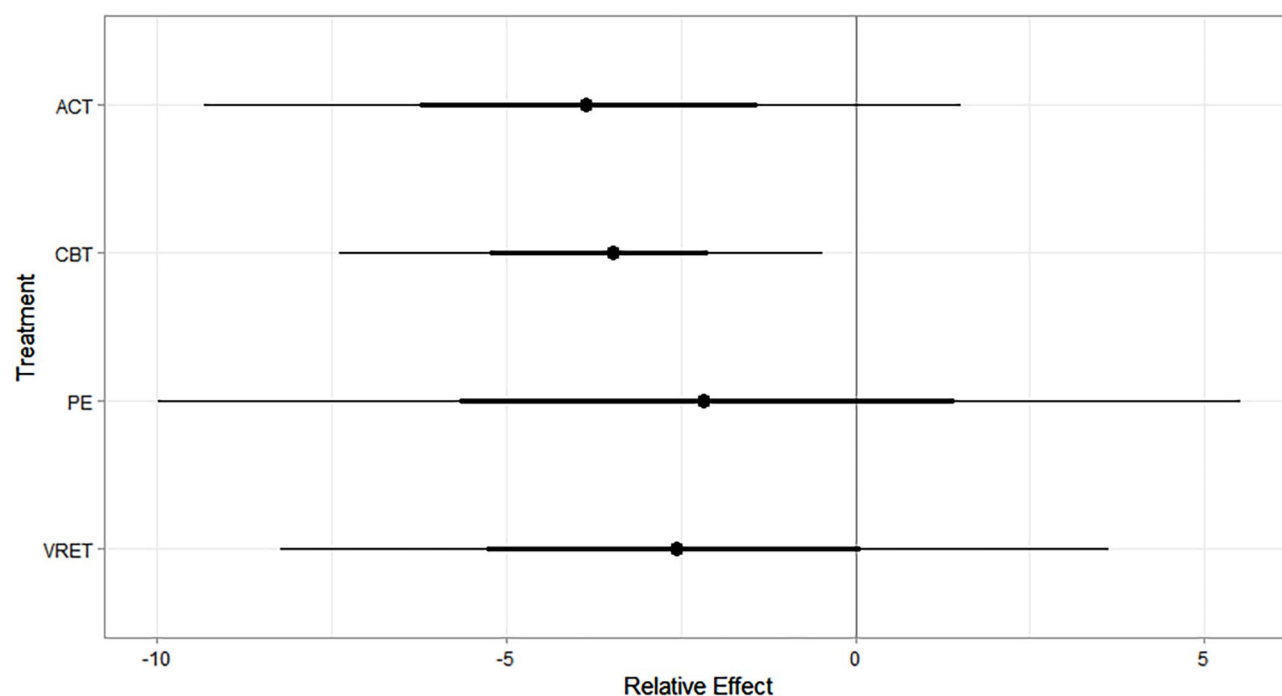


Fig. 4 Relative effects plot

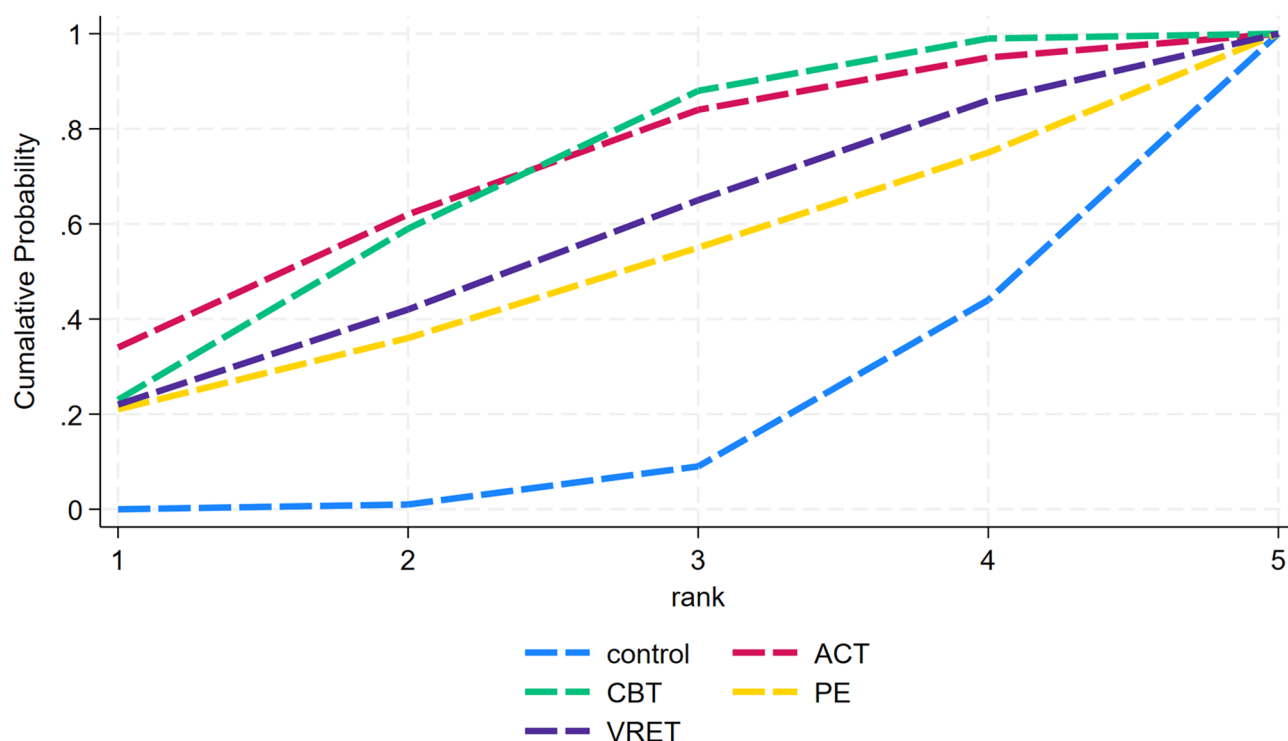


Fig. 5 SUCRA plot

supports the effectiveness of CBT, which, in line with the majority of research, significantly alleviates anxiety disorders in children and adolescents [99–101]. A recent systematic review and further demonstrated CBT's efficacy in treating adolescent anxiety and in reducing relapse rates, including for younger children when appropriately adapted [19].

There is still a lack of systematic literature search reviews on the effects of VRET on anxiety disorders in adolescents specifically targeting children and adolescents, and its great potential in the treatment of anxiety disorders in children and adolescent contrasts with the considerable lack of controlled trials [102]. In our analysis of the comparative efficacy of various interventions for adolescent anxiety, we included six randomized controlled trials on VRET. Our results suggest that VRET is effective in the treatment of anxiety disorders in adolescents, with efficacy second only to psychotherapies such as ACT and CBT. The fundamental principle of VRET is to create a virtual environment that simulates emotional network processing in the brain. When a patient encounters a threatening stimulus within this virtual environment, it activates the brain's fear network, triggering a fear response [103]. As new, non-threatening information is gradually introduced, the fear structure of the patient is modified through habituation, making the stimulus less threatening. This exposure continues until the patient's anxiety and fear decrease to a level where therapeutic effects are achieved [104, 105]. Additionally,

Virtual reality has been shown to effectively reduce pain and anxiety in adolescents undergoing various medical procedures, such as burn care, post-burn physical therapy, dental treatments, and needle-related surgeries, with a stronger effect compared to adults [106, 107].

It is not surprising to see PE ranked in Relative effects and SUCRA. While its efficacy is lower than that of ACT, CBT, and VRET, the difference is not much, suggesting that PE may produce effects comparable to psychotherapy and VRET. Several studies support this, also highlighting the benefits of PE in terms of cost, side effects, and additional health advantages [108]. Moreover, when included as part of a combined therapeutic approach for anxiety, PE provides added effectiveness [109]. The effects of PE in the treatment of anxiety disorders may be related to several mediators, including reduction of pain intolerance, modulation of neurotransmitter activity (e.g., serotonin, norepinephrine, gamma-aminobutyric acid [GABA]), increase in brain-derived neurotrophic factor (BDNF), and behavioral activation effects and stress response effects [110, 111]. While our findings support the potential use of PE in treating anxiety disorders, the existing evidence remains limited, with many studies having low overall quality [112]. Therefore, further research is needed to explore the impact of PE, particularly in clinical populations of adolescents with anxiety disorders.

Furthermore, no studies meeting the inclusion criteria for combination therapies were identified during the

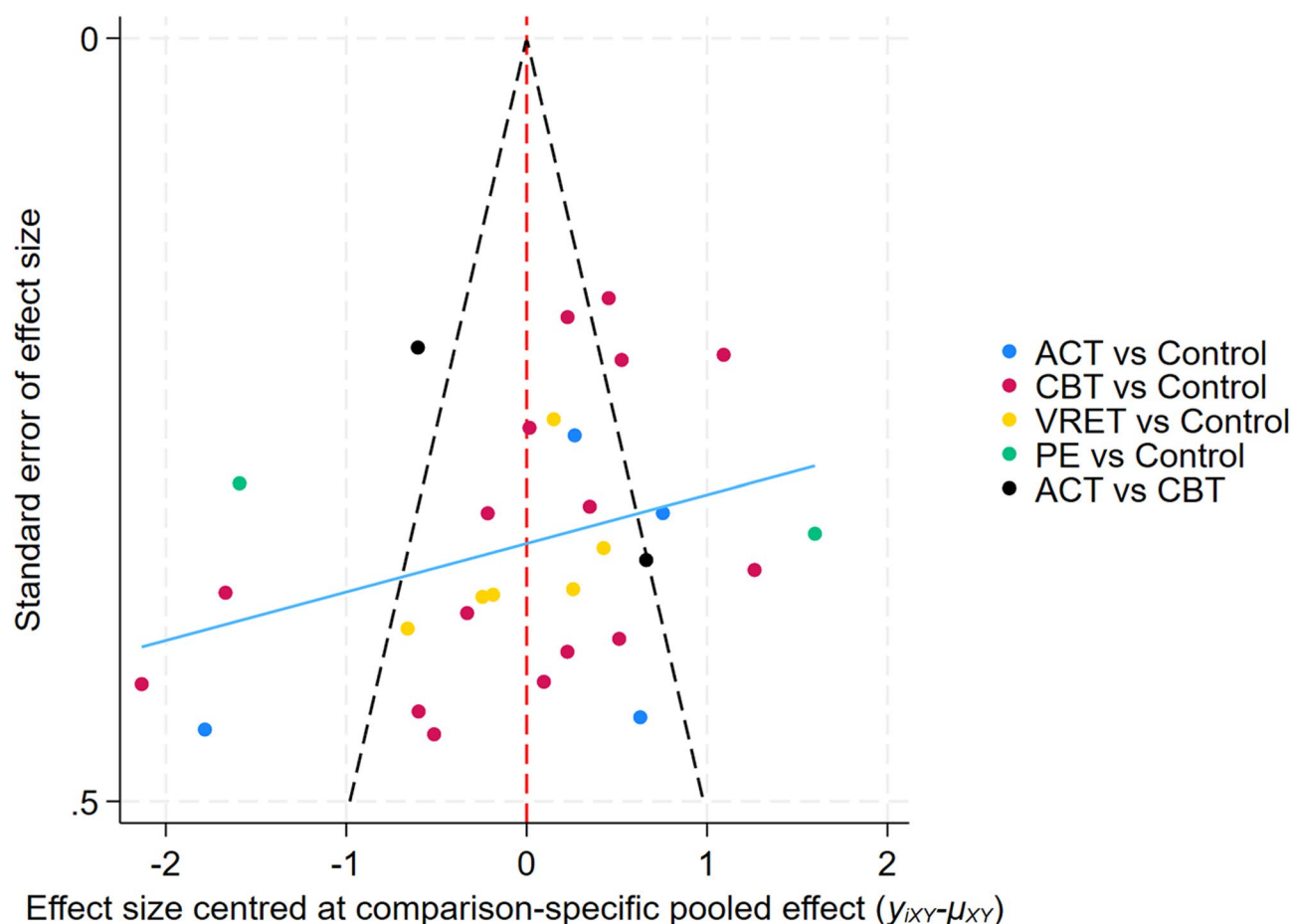


Fig. 6 Funnel plot

literature screening. However, the potential of combination therapies in treating anxiety disorders has been explored. Research has shown that incorporating PE into a comprehensive treatment plan can improve outcomes [109]. For example, Merom et al. [109] found that 8 to 10 weeks of group CBT combined with exercise significantly reduced anxiety symptoms compared to lifestyle education groups. Additionally, Bouchard et al. [114] demonstrated that combining virtual reality with CBT was more effective than standard CBT alone. A meta-analysis by Wu et al. [103] further confirmed that the VR-CBT combination provided similar therapeutic effects to standard CBT, while offering more timely intervention for anxiety disorders. While standalone virtual reality and PE may not fully address the cognitive needs required for effective psychological treatment, they each offer distinct advantages. PE, as an accessible and affordable treatment, can be widely disseminated, addressing barriers such as treatment access, affordability, and stigma [109]. Likewise, as virtual reality technology evolves, it becomes easier to implement and more cost-effective compared to traditional treatments [103]. Moreover, the anonymity and privacy offered by virtual reality environments

may encourage greater participation, as patients need not worry about potential embarrassment. Given these advantages, combining PE or VR-assisted therapy could be a promising approach for treating adolescent anxiety disorders. Future research should focus on evaluating the effectiveness of these combination therapies.

Strengths and limitations

This meta-analysis has several strengths. First, the study protocols were preregistered in the PROSPERO database, ensuring transparency and adherence to established research standards. Second, a comprehensive systematic evaluation was conducted following the PRISMA guidelines, which enhanced the scientific rigor and reliability of the included studies. Third, to minimize bias and ensure validity, the search, data extraction, and quality assessment processes were independently carried out by two researchers. Finally, this study utilized the Bayesian network meta-analysis, integrating both direct and indirect evidence to compare the most effective interventions for adolescent anxiety disorders.

This meta-analysis has several limitations. First, although Egger's test did not indicate a small-study

effect, potential publication bias cannot be ruled out. This includes language bias, as only English-language studies were included, and accessibility bias due to the exclusion of gray literature. Second, many of the included trials did not use blinding and provided insufficient information on allocation concealment, increasing the risk of performance and selection bias. These methodological weaknesses may undermine the internal validity and reduce confidence in the estimated effects. Third, although all included studies were randomized controlled trials, the overall certainty of the evidence was rated as low based on the GRADE approach. This was primarily due to a high risk of bias in several studies and substantial imprecision in treatment effect estimates. Finally, substantial heterogeneity across studies resulted in wide credible intervals. For most interventions—such as ACT, PE, and VRET—the intervals included the possibility of null effect, with only CBT showing consistently positive outcomes. Furthermore, the 95% credible intervals for treatment rankings showed considerable overlap, which reduces confidence in identifying ACT as the most effective intervention. Considering these limitations, this Bayesian network meta-analysis should be regarded as hypothesis generating, and its findings interpreted with caution.

Conclusion

This review used a Bayesian network meta-analysis to compare the effectiveness of various interventions for treating anxiety disorders in children and adolescents. The findings suggest that ACT may be the most effective intervention among those evaluated. Clinicians may consider incorporating ACT into treatment plans where available, while also taking into account factors such as accessibility, patient preferences, and comorbid conditions when selecting interventions. However, these findings should be interpreted with caution due to the overall low quality of evidence, substantial heterogeneity, and imprecision in the results. Future research should investigate the potential benefits of combination therapies involving PE or VR-assisted treatments. Additionally, large-scale, high-quality randomized controlled trials are needed to further validate or refine these findings.

Abbreviations

ACT	Acceptance and Commitment Therapy
CBT	Cognitive Behavioral Therapy
VR	Virtual Reality
VRET	Virtual Reality Exposure Therapy
PE	Physical Exercise
DSM	Diagnostic and Statistical Manual of Mental Disorders
MD	Mean Difference
CrI	Credibility Intervals
SUCRA	Surfaces Under the Cumulative Ranking Area
RC	Randomized Controlled Trials

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s12888-025-07227-y>.

Supplementary Material 1

Supplementary Material 2

Acknowledgements

Not applicable.

Author contributions

LLH: Conceptualization, Funding acquisition, Data curation, Formal analysis, Methodology, Writing – original draft. LQE: Methodology, Formal analysis, Writing – original draft. WJY: Methodology, Formal analysis, Data curation, Writing – original draft. FG: Conceptualization, Funding acquisition, Supervision, Writing – original draft, Writing– review & editing. MC: Conceptualization, Supervision, Writing – original draft, Writing – review & editing.

Funding

This work was supported by the [China Association for Science and Technology Graduate Student Science Popularization Ability Improvement Project#1] under Grant [number KXYJS2024007], [Beijing Municipal University Great Wall Scholar Training Program Project#2] under Grant [number CIT&TCD20180335] and [Emerging Interdisciplinary Platform for Medicine and Engineering in Sports (EIPMES)#3].

Data availability

Data is provided within the manuscript or supplementary information files.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

Clinical trial number.

Not applicable.

Appendices

Appendix A. PRISMA checklist.

Appendix B. Supplementary Materials.

Received: 17 February 2025 / Accepted: 21 July 2025

Published online: 26 August 2025

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